

AI/ML Internship



Task: 01

TOPIC: Python Programming, Data Analysis
& Visualization

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Internship Domain: Artificial Intelligence &
Machine Learning

Dataset Used: Netflix Movies and TV Shows

Submission Date: (20-07-2026)

OBJECTIVES

The objective of this project is to understand the basics of Python programming and data analysis by loading a real-world dataset, cleaning the data, performing exploratory data analysis (EDA), creating visualizations, and drawing meaningful insights.

Dataset Description

The dataset used in this project is the **Netflix Movies and TV Shows Dataset**, downloaded from Kaggle. It contains information about movies and TV shows available on Netflix.

Dataset Details

- Dataset Name: Netflix Movies and TV Shows
- Source: Kaggle
- Total Records: 8807
- Total Columns: 12

The dataset contains **information such as:**

show_id

- . type
- . title
- . director
- . cast
- . country
- . date_added
- . release_year
- . rating
- duration
- . listed_in
- . description

Methodology:

The project was completed using the following steps:

1. Data Loading

- Imported Python libraries.
- Loaded the dataset using Pandas.
- Displayed the first and last five records.
- Checked dataset information.

2. Data Cleaning

- Checked missing values.
- Filled missing values.
- Removed duplicate records.
- Converted incorrect data types.
- Renamed columns where necessary.

3. Exploratory Data Analysis

Performed statistical analysis including:

- . Mean
- . Median
- . Mode
- . Standard Deviation
- . Correlation Matrix

4. Data Visualization

Created multiple charts including:

- . Histogram
- . Bar Chart
- . Pie Chart
- . Scatter Plot
- . Heatmap
- . Box Plot

- Count Plot
- Pair Plot
- Line Chart
- Violin Plot

5. Insight

- Movies are significantly more numerous than TV Shows.
- Most content was released after 2015.
- TV-MA is the most common rating.
- The dataset contains content from many countries.
- Some directors have produced multiple Netflix titles.
- There are only a few missing values after cleaning.

- . Duplicate records were removed successfully.
- . The release year has a wide distribution.
- . The correlation matrix shows very weak relationships because there are few numeric variables.
- . Netflix has rapidly increased its content library in recent years.

SCREENSHOTS

1) Import Libraries

The screenshot shows a Jupyter Notebook interface with the following components:

- Browser Address Bar:** `colab.research.google.com/drive/1V0Z4LRhyv--PshEwuiSPxs4NwUqc15m#scrollTo=WvdwZe-ym-u7`
- File Explorer:** Shows the notebook file `Netfix_EDA.ipynb`.
- Code Cell:** Contains the following Python code:

```
import pandas as pd
import numpy as np

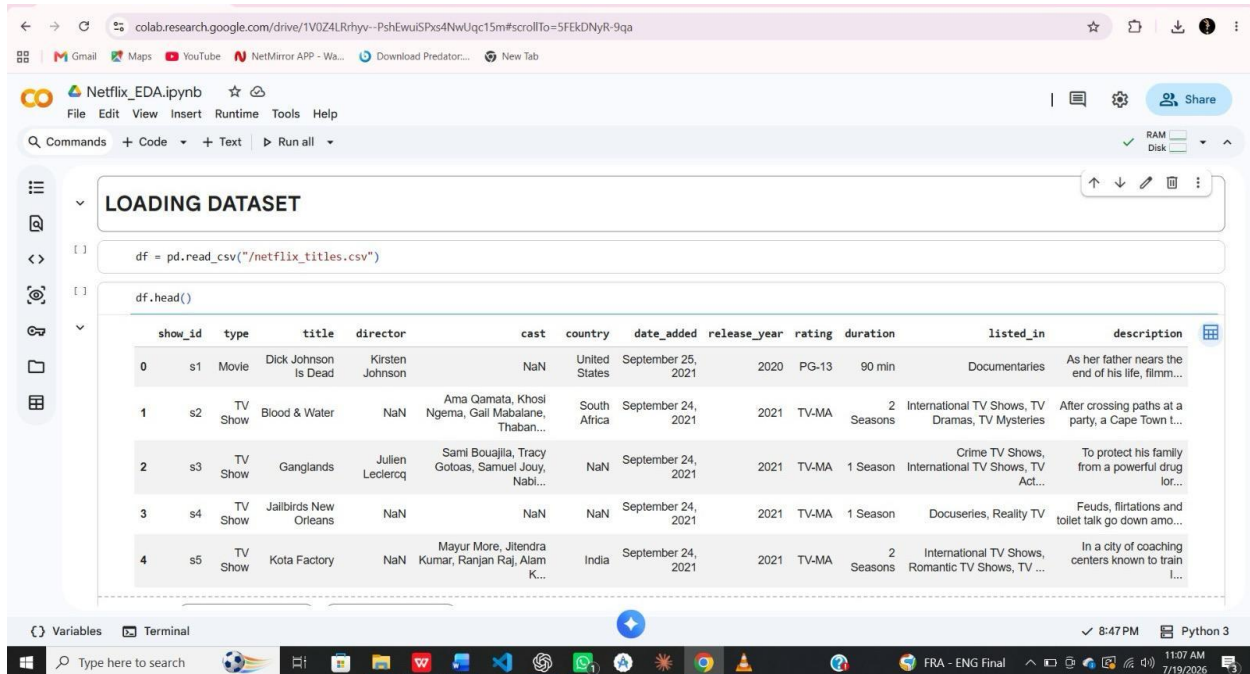
import matplotlib.pyplot as plt
import seaborn as sns

# Display all columns
pd.set_option("display.max_columns", None)

# Graph Style
sns.set(style="whitegrid")
```
- Output Cell:** Displays the first row of a DataFrame using `df.head()`.

show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
- Bottom Bar:** Shows the status bar with "Variables", "Terminal", "8:47 PM", and "Python 3".

2)Loading Dataset



The screenshot shows a Google Colab notebook interface. The browser address bar indicates the notebook is located at a Google Drive link. The notebook title is "Netflix_EDA.ipynb". The menu bar includes File, Edit, View, Insert, Runtime, Tools, and Help. The toolbar shows a search bar, a "Run all" button, and a RAM/Disk usage indicator.

The notebook has two cells. The first cell contains the code to load a CSV file:

```
df = pd.read_csv("/netflix_titles.csv")
```

The second cell contains the code to display the first five rows of the dataset:

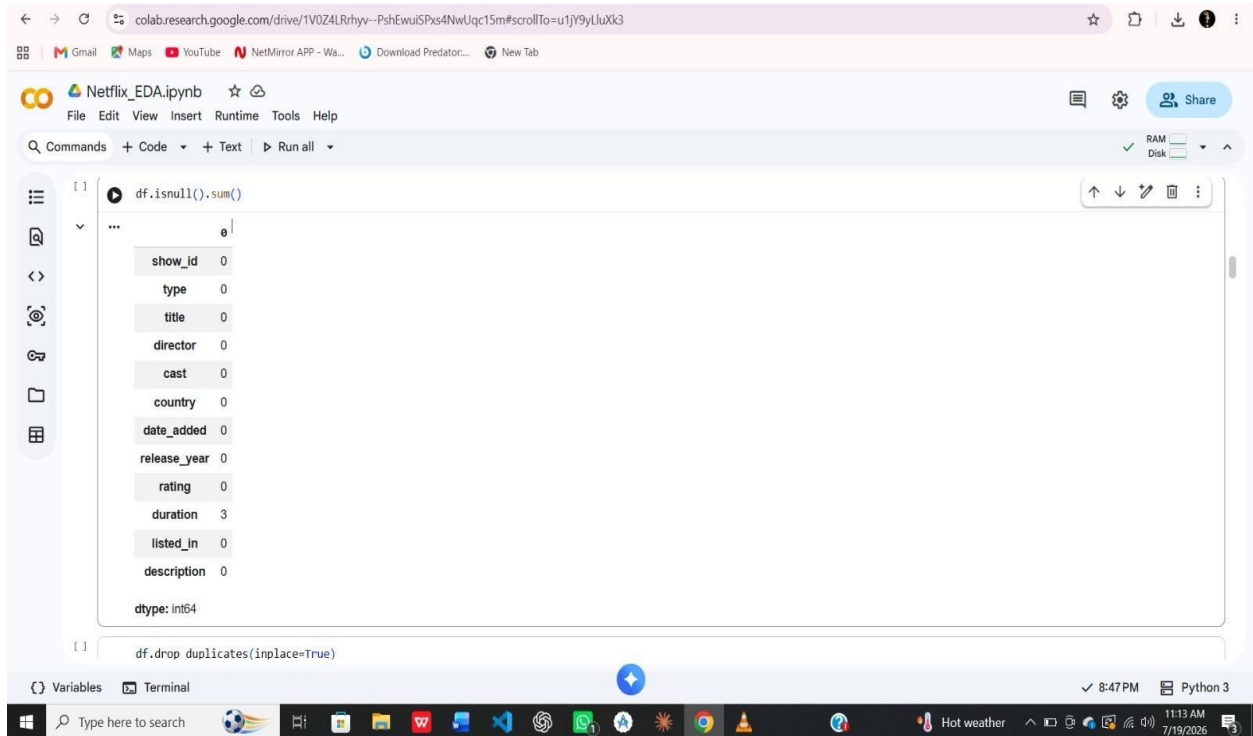
```
df.head()
```

The output of the second cell is a table with 13 columns: show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, and description. The table contains 5 rows of data.

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Oamata, Khosi Ngema, Gail Mablane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town L...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabl...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

The bottom of the notebook shows a "Variables" tab with a "Terminal" button. The system tray at the bottom indicates the time is 8:47 PM and the user is using Python 3.

3) Cleaning Data



The screenshot shows a Jupyter Notebook interface with the following code and output:

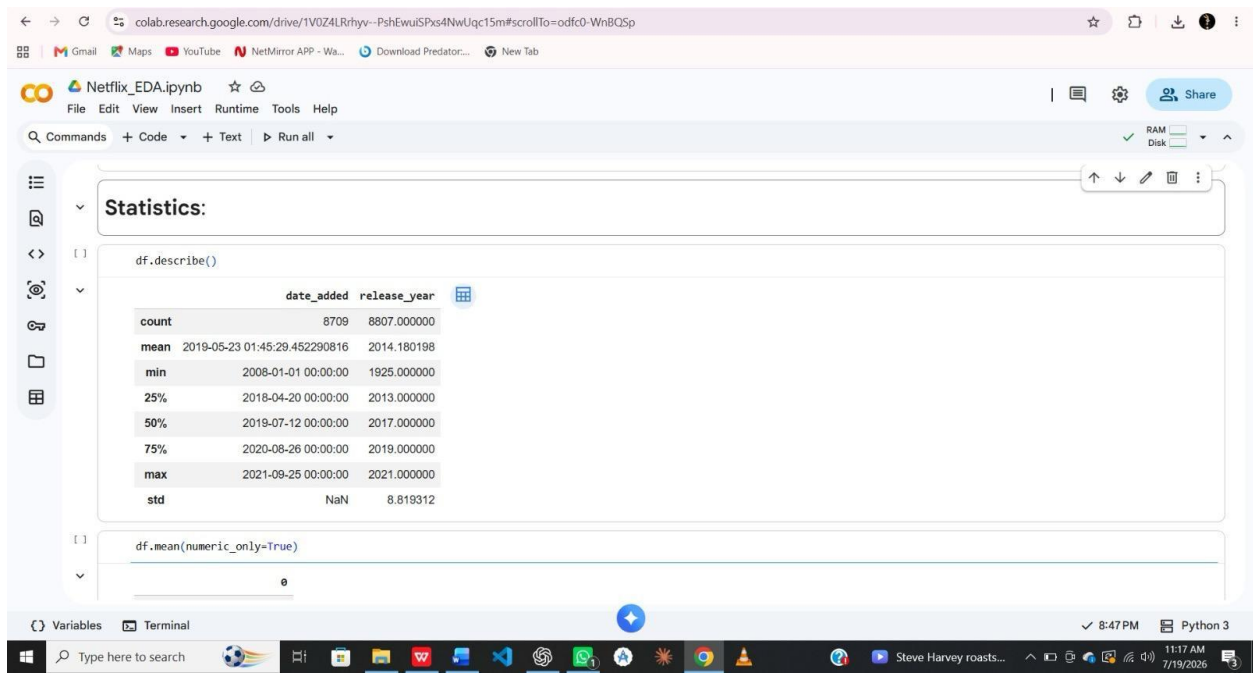
```
[1]: df.isnull().sum()
```

show_id	0
type	0
title	0
director	0
cast	0
country	0
date_added	0
release_year	0
rating	0
duration	3
listed_in	0
description	0
dtype:	int64

```
[2]: df.drop_duplicates(inplace=True)
```

The interface includes a top navigation bar, a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), and a toolbar with icons for variables, terminal, and search. The status bar at the bottom shows the time as 8:47 PM and the Python version as Python 3.

4) Visualization Code



The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[1]: df.describe()
```

Statistics:

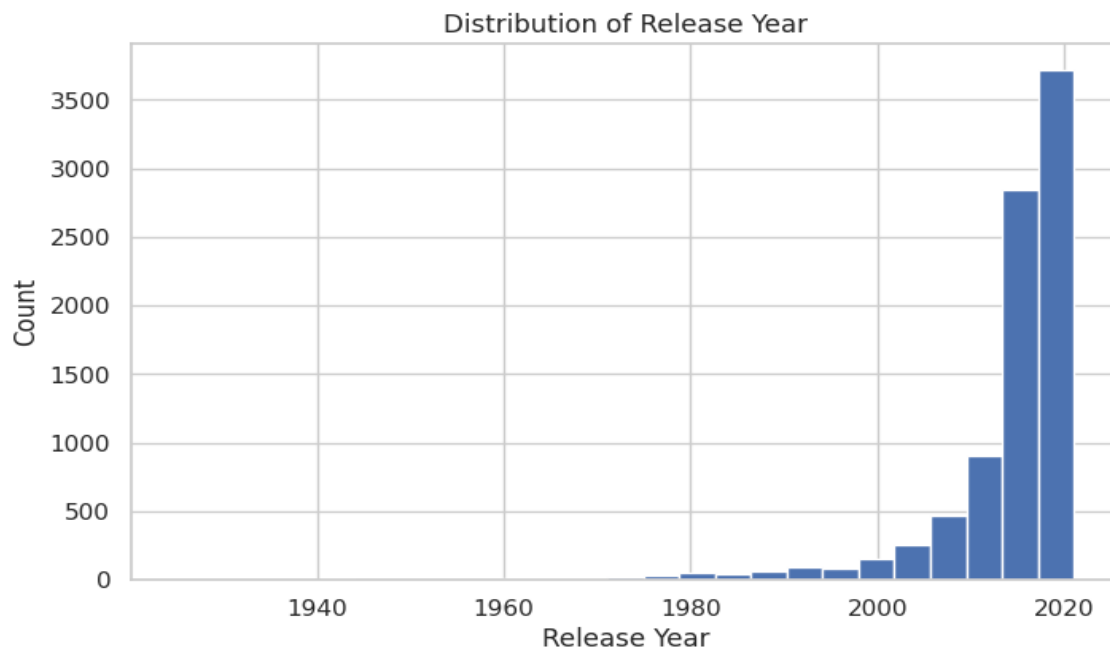
	date_added	release_year
count	8709	8807.000000
mean	2019-05-23 01:45:29.452290816	2014.180198
min	2008-01-01 00:00:00	1925.000000
25%	2018-04-20 00:00:00	2013.000000
50%	2019-07-12 00:00:00	2017.000000
75%	2020-08-26 00:00:00	2019.000000
max	2021-09-25 00:00:00	2021.000000
std	NaN	8.819312

```
[2]: df.mean(numeric_only=True)
```

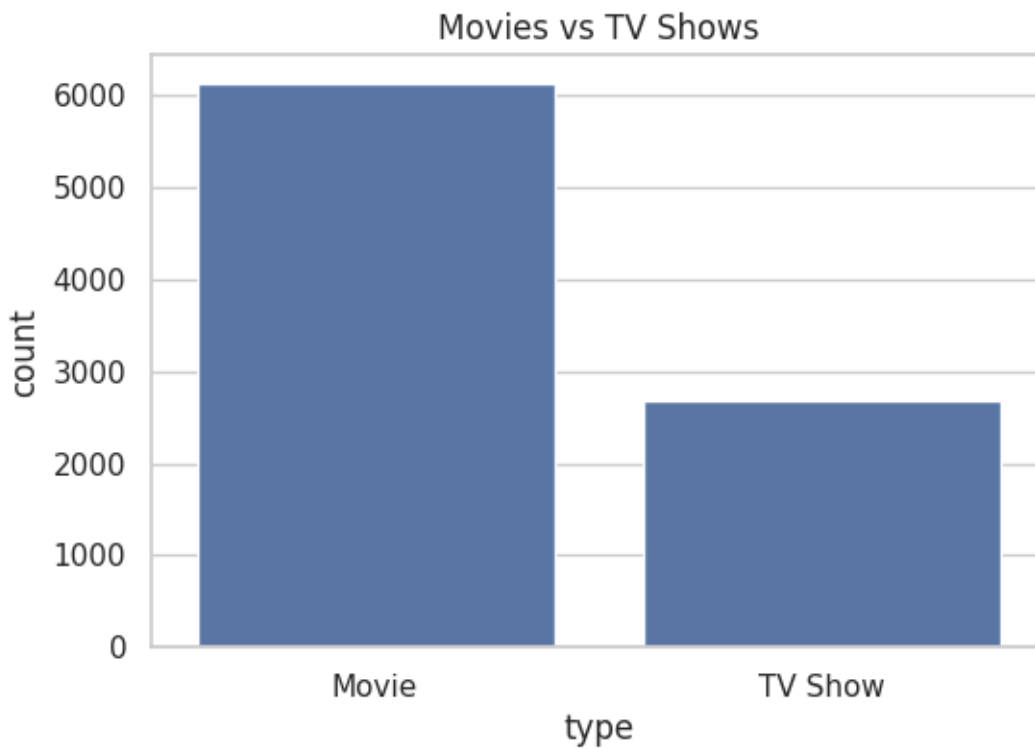
The output of the second code cell is a single value: 0.

The interface includes a top navigation bar, a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), and a toolbar with icons for variables, terminal, and search. The status bar at the bottom shows the time as 8:47 PM and the Python version as Python 3.

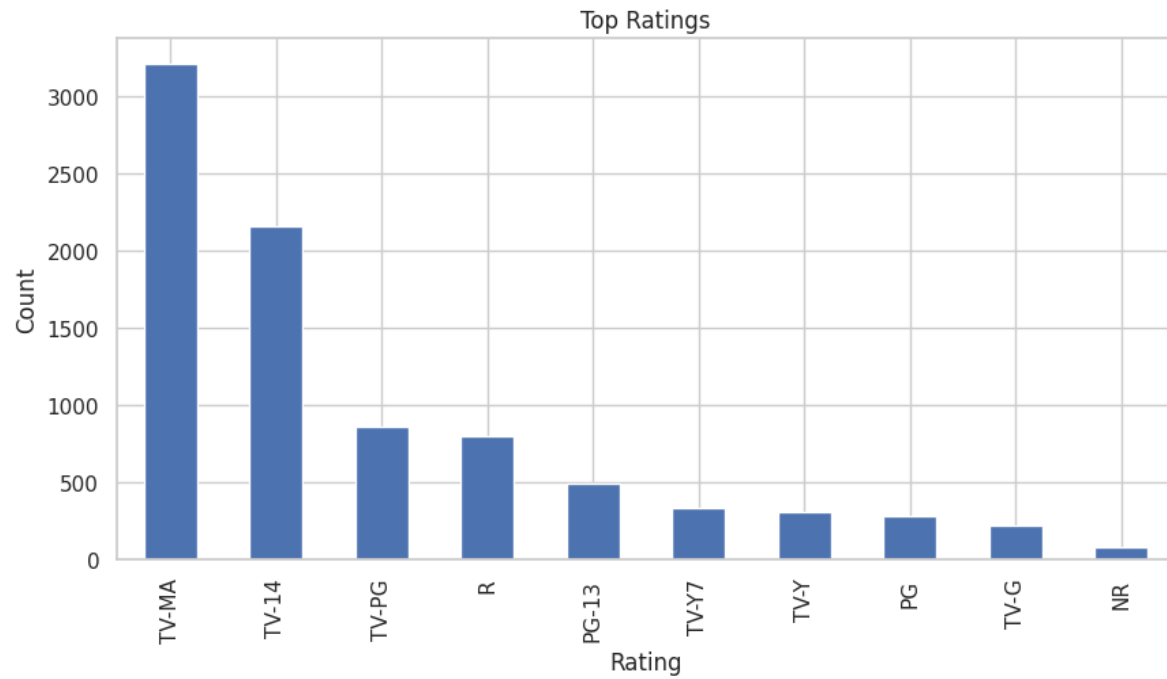
GRAPHS:1) Histogram



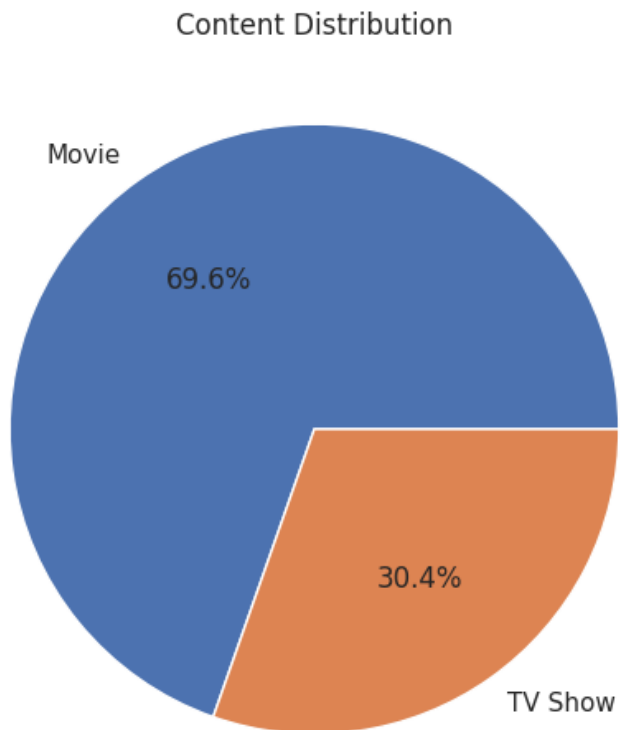
2) Count Plot



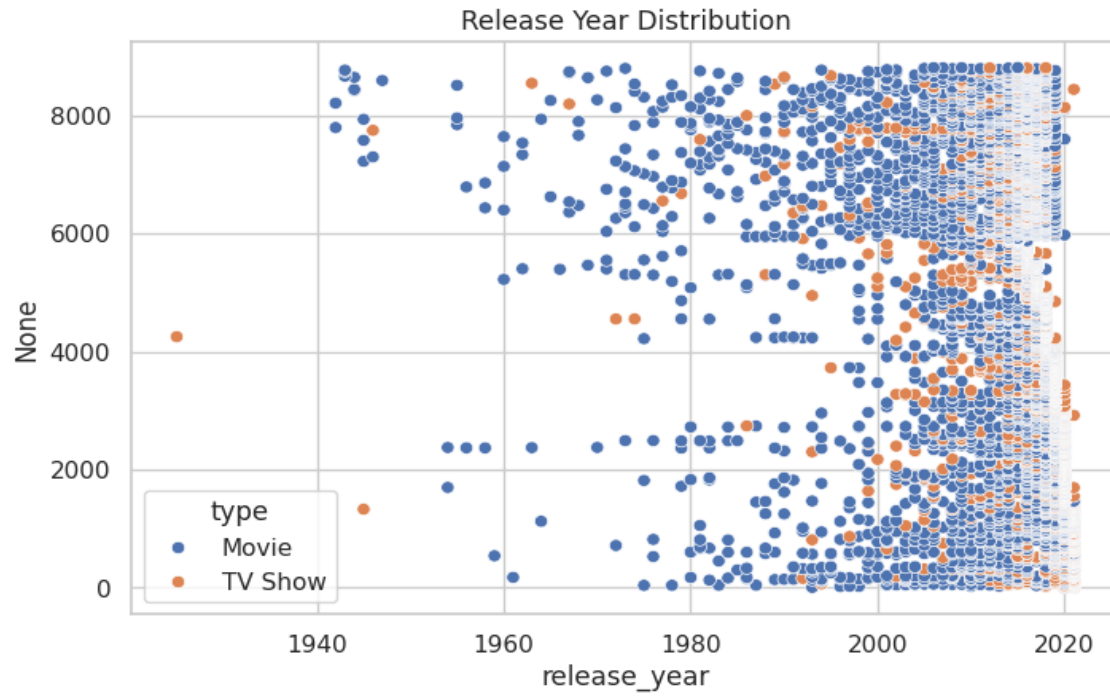
3) Bar Chart



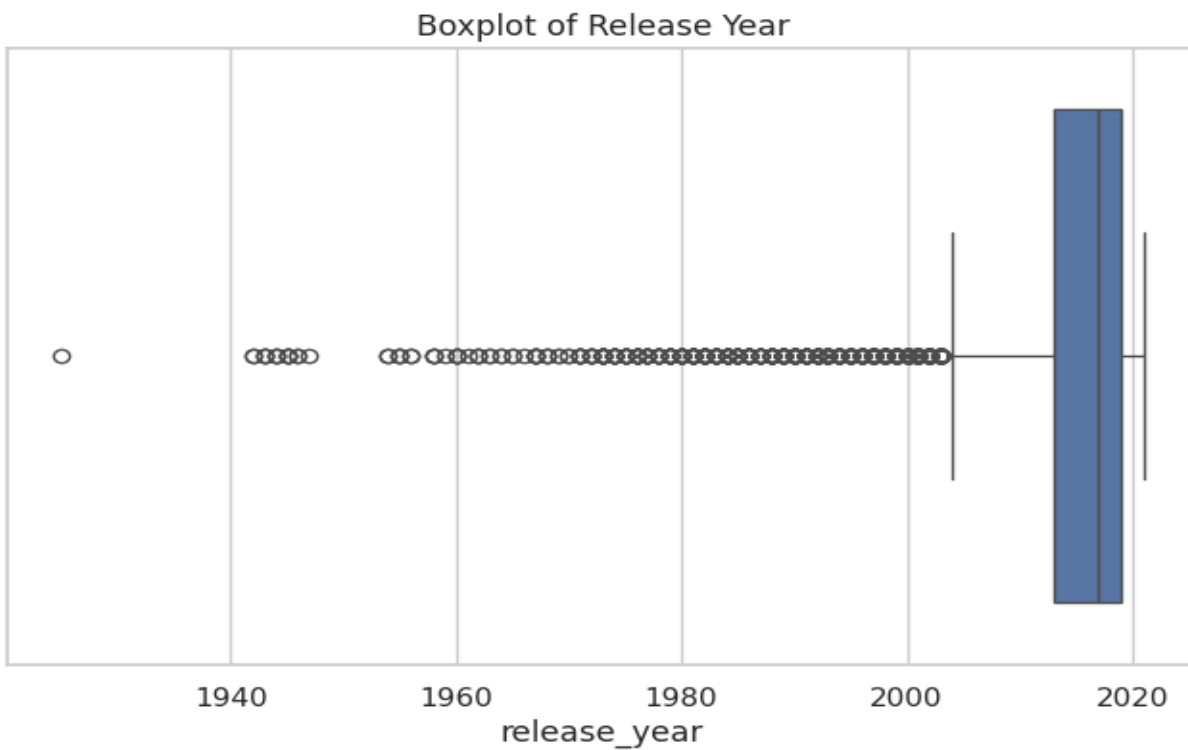
4) Pie Chart



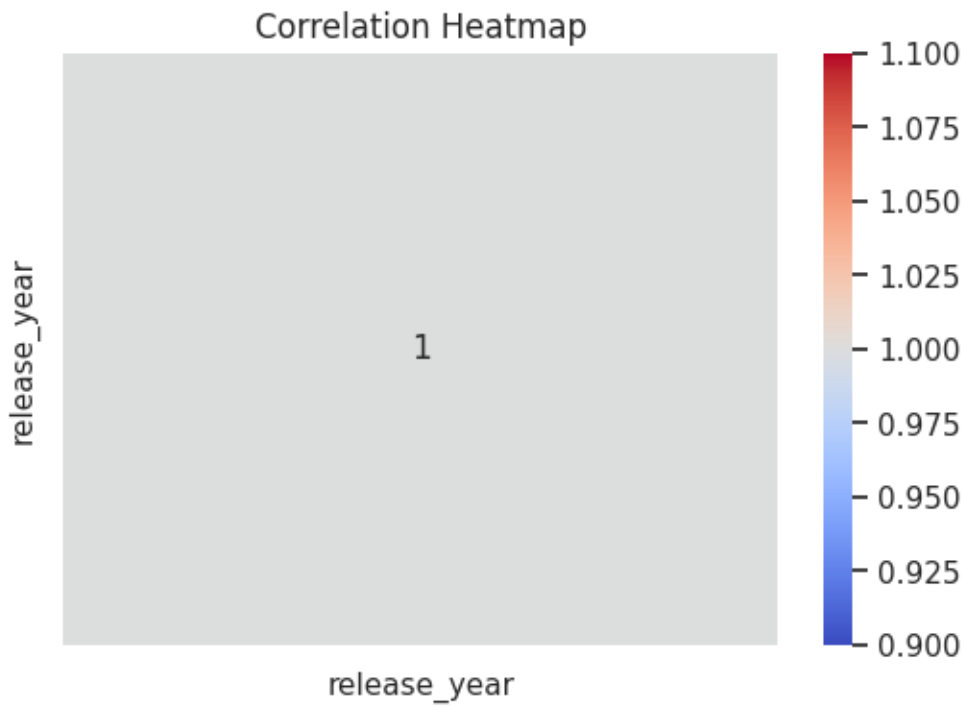
5) Scatter Plot



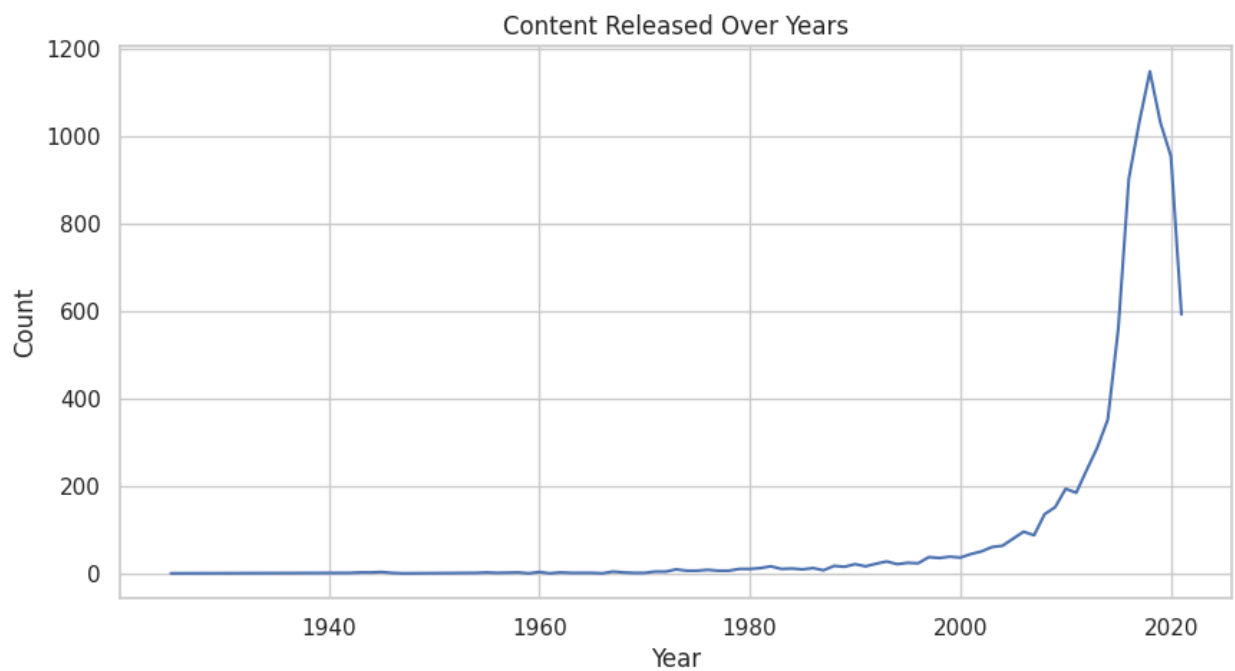
6) Box Plot



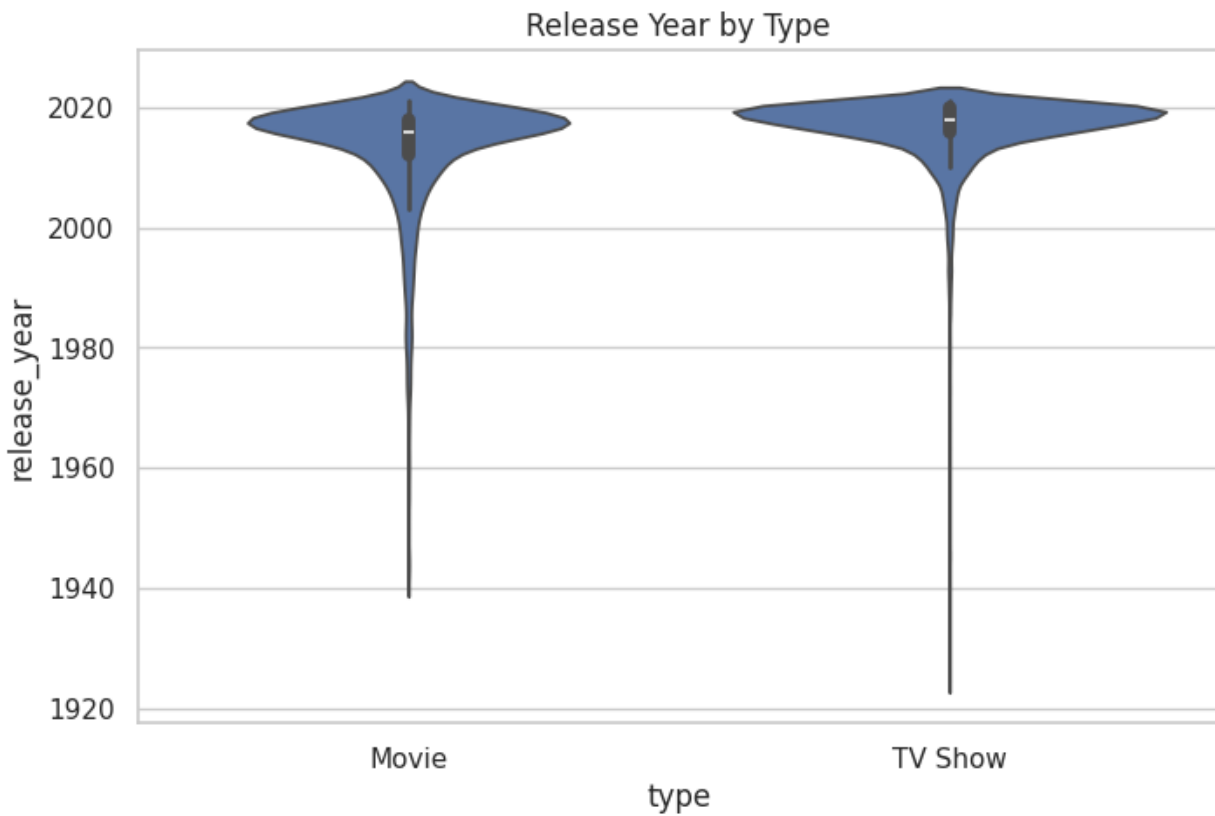
7) Heatmap



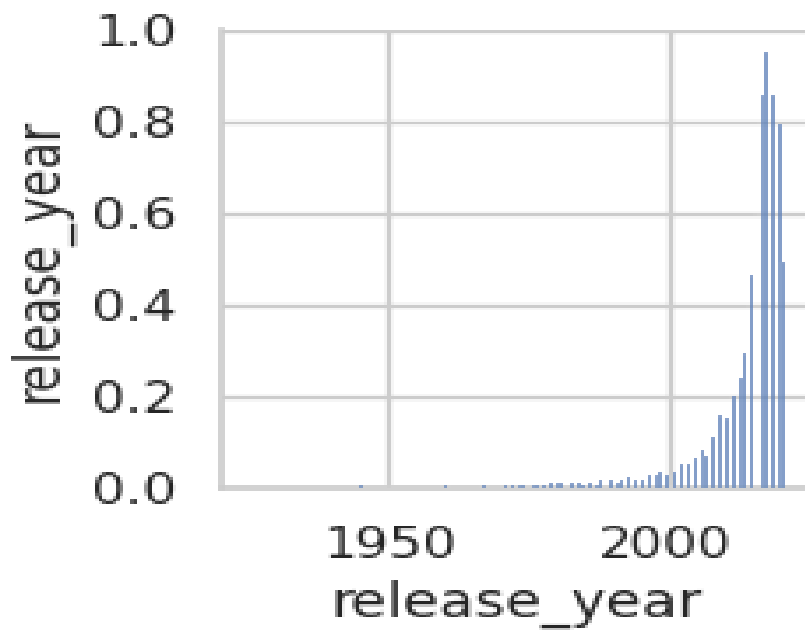
8) Line Chart



9) Violin Plot



10) Pair Plot



Results:

After cleaning the dataset, all missing values were handled successfully.

Duplicate records were removed, and the dataset was prepared for analysis.

Statistical analysis provided useful information about the numerical columns.

Ten different visualizations were created to understand the distribution of content, ratings, release years, and other important characteristics.

The project successfully demonstrated the complete data analysis workflow using Python.

Observations:

- Movies are more common than TV Shows.
- Most Netflix content was released after 2015.
- TV-MA is the most frequent rating.
- The USA has the highest number of Netflix titles.
- Many directors have only one title listed.
- Missing values were successfully handled.
- Duplicate records were removed.

- Most content belongs to Drama and International Movies.
- Release year shows a growing trend in recent years.
- Numerical features have weak correlation.

Conclusion:

This project provided practical experience with Python programming and data analysis techniques. The Netflix dataset was successfully cleaned and analyzed using Pandas, NumPy, Matplotlib, and Seaborn. Various visualizations helped identify important trends and patterns. Overall, this task improved understanding of data preprocessing, exploratory data analysis, and visualization.

